

3.6 Binomial Distribution

Given is a **vase** with **a red and b white balls**. Someone selects one ball from this vase. If this ball is red, then it is a **"success"**. In the case of a white ball, there is a **"failure"**.

Such an experiment is called a **Bernoulli trial with constant probability of success $p (= \frac{a}{a+b})$** .

A **Bernoulli trial** can be described by a discrete random variable X :

$$X = \begin{cases} 0 & , \text{ in case of failure} \\ 1 & , \text{ in case of success} \end{cases}$$

So

Table 3.2 The probability distribution of a Bernoulli trial

x	0	1
$P(X = x)$	$1 - p$	p

We call X an **alternative random variable with parameter p** .

Remarks

- Notation: **$X \sim \text{alt}(p)$**
- $E(X) = 0 \cdot (1 - p) + 1 \cdot p = p$
- $V(X) = E(X^2) - E(X)^2$
 $= 0^2 \cdot (1 - p) + 1^2 \cdot p - p^2 = p - p^2 = p \cdot (1 - p)$

Sampling With Replacement

Someone selects n objects **with replacement** from a set that contains a red and b white objects. So we have n independent **Bernoulli trials** with probability of success $p = \frac{a}{a+b}$.

Let the random variable X be the number of successes (i.e. red objects) in the sample.

Then

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

The random variable X is a **binomial random variable with parameters n and p** .

Remarks

- $\binom{n}{x}$ is called the **binomial coefficient** n over x : $\binom{n}{x} = \frac{n!}{x!(n-x)!}$.
- Using the **graphical calculator**¹ ([gc/math/prob/option 3](#)): $\binom{n}{x} = n \text{ nCr } x$.
- Notation: $X \sim \text{bin}(n;p)$.
- If $X = X_1 + X_2 + \dots + X_n$, with $X_i \sim \text{alt}(p)$ and mutually independent, then

$$X \sim \text{bin}(n;p).$$

So

$$\begin{aligned} E(X) &= E(X_1 + X_2 + \dots + X_n) \\ &= E(X_1) + E(X_2) + \dots + E(X_n) = n \cdot p \end{aligned}$$

- Analogously and using mutual independency (see [Chapter 5](#)):

$$\begin{aligned} V(X) &= V(X_1 + X_2 + \dots + X_n) \\ &= V(X_1) + V(X_2) + \dots + V(X_n) = n \cdot p \cdot (1 - p) \end{aligned}$$

Summarizing

$E(X)$	$=$	$n \cdot p$
$V(X)$	$=$	$n \cdot p \cdot (1 - p)$

Example 7

Let X be a binomial random variable with $n = 15$ and $p = 0.4$. Then

- $P(X = 10) = \begin{cases} \text{binompdf}(15, 0.4, 10) &= 0.0245 & \text{gc/distr/option a} \\ 15 \text{ nCr } 10 \cdot 0.4^{10} \cdot 0.6^5 &= 0.0245 & \text{gc/math/prob/3} \\ P(X \leq 10) - P(X \leq 9) &= 0.9907 - 0.9662 & \text{Table II} \\ &= 0.0245 \end{cases}$
- $P(X \geq 10) = 1 - P(X \leq 9)$
 $= \begin{cases} 1 - \text{binomcdf}(15, 0.4, 9) &= 0.0338 & \text{gc/distr/b} \\ 1 - 0.9662 &= 0.0338 & \text{Table II} \end{cases}$

¹ TI-84 Plus CE-T

3.7.1 Geometric Distribution

Assume a series of **Bernoulli trials** (i.e. independent trials with constant probability p of a success on each trial. Trials are conducted until a success is obtained. Let the random variable X denote the number of trials until the first success. Then

$$P(X = x) = (1 - p)^{x-1} \cdot p \quad \text{with } x = 1, 2, 3, \dots$$

The random variable X is a geometric random variable with parameter $0 < p < 1$.

Remarks

- Notation: $X \sim \text{geo}(p)$
- $E(X) = \frac{1}{p}$
- $V(X) = \frac{1-p}{p^2}$

Example 8

It takes an average of five attempts to achieve a success. What is the probability of success in no more than three times?

So determine $P(X \leq 3)$ with $X \sim \text{geo}(p = 0.2)$.

- $$\begin{aligned} P(X \leq 3) &= P(X = 1) + P(X = 2) + P(X = 3) \\ &= 0.2 + (1 - 0.2) \cdot 0.2 + (1 - 0.2)^2 \cdot 0.2 \\ &= 0.488 \end{aligned}$$
- Using the graphical calculator (**gc/distr/g**):
$$\begin{aligned} P(X \leq 3) &= \text{geometcdf}(.2, 3) \\ &= 0.488 \end{aligned}$$

Siméon Poisson



3.9 Poisson Distribution

The Frenchman **Poisson (1781 – 1840)** derived the following formula for the probability of rare events:

$$P(X = x) = e^{-\lambda} \cdot \frac{\lambda^x}{x!}, \text{ where } x = 0, 1, 2, \dots$$

with X = the number of rare events and λ = the average number of this event.

The random variable X is a Poisson random variable with parameter $\lambda > 0$.

The following example shows that the Poisson formula is a plausible formula.

Example 9 **London (1944)**

Dividing the map of London into 24 by 24 (= 576) sectors and then counting how many sectors count 0, 1, 2, 3, 4 or 5 or more direct hits by V2 rockets, the data shown in Table 3.3 will result.

Table 3.2 The number of hits by V2 rockets

Number of hits	Observed number of sectors	Expected number of sectors
0	229	227.5
1	211	211.3
2	93	98.1
3	35	30.4
4	7	7.1
≥ 5	1	1.5

So the average number of direct hits in a sector is equal to

$$\lambda = \frac{229 \cdot 0 + 211 \cdot 1 + 93 \cdot 2 + 35 \cdot 3 + 7 \cdot 4 + 1 \cdot 5}{576} = 0.9288$$

Thus

$$P(X = 1) = e^{-0.9288} \cdot \frac{0.9288^1}{1!} = 0.3669$$

So it follows

$$\begin{array}{llll} \text{expected number} & = & 0.3669 \cdot 576 & = & 211.3; \\ \text{observed number} & & & = & 211. \end{array}$$

Thus the difference between theory and reality is small.

Remarks

- Notation: $X \sim \text{Poi}(\lambda)$
- The Poisson distribution is used in various situations:
 - the number of machine failures;
 - the number of ships entering a port;
 - the number of goals in a soccer match.
- The following formulas apply to the expected value and the variance of a Poisson distributed random variable X :

$$\begin{array}{rclcl} \mu & = & E(X) & = & \lambda \\ \sigma^2 & = & V(X) & = & \lambda \end{array}$$

Example 10

Let X be the number of customers visiting a post office every 15 minutes. Suppose $X \sim \text{Poi}(\lambda = 3/15 \text{ minutes})$.

- Determine $P(X = 4)$.

- Using the **Poisson formula**

$$P(X = 4) = e^{-3} \cdot \frac{3^4}{4!} = \frac{81}{24} e^{-3} = 0.1680$$

- Using the graphical calculator (**gc/distr/d**):

$$P(X = 4) = \text{poissonpdf}(3, 4) = 0.1680$$

or:

$$\begin{aligned} P(X = 4) &= P(X \leq 4) - P(X \leq 3) \\ &= \text{poissoncdf}(3, 4) - \text{poissoncdf}(3, 3) \\ &= 0.8153 - 0.6472 = 0.1680 \end{aligned}$$

- Let Y be the number of customers visiting the post office every hour. Determine $P(Y = 4)$.

Since $Y \sim \text{Poi}(\lambda = 4 \cdot 3 = 12)$, it follows:

$$P(Y = 4) = \text{poissonpdf}(12, 4) = 0.0053$$

Theorem

$$\left. \begin{array}{l} X \sim \text{Poi}(\lambda_1) \\ Y \sim \text{Poi}(\lambda_2) \\ X, Y \text{ independent} \end{array} \right\} \Rightarrow X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$$